# EndNote MCP: read-only access to EndNote 2025 libraries

*Claude Code can search, read and check your EndNote libraries. It never changes a library, a .Data folder or an attachment.*

## Set up

- **Registered already.** The server is in your user config as **endnote**. Check it with **claude mcp list** (it should show Connected).
- Libraries are listed in **libraries.json** next to server.py. Add a library there with its .enl path, then start a new Claude Code session so the list refreshes.
- **EndNote can stay open.** The server copies a library only after EndNote has saved it. If a copy fails, the server shows the last copy with its time.

## Tools

| Tool | What it does |
|---|---|
| `list_libraries` | Configured libraries, reference counts, snapshot time. |
| `search_items` | Words in author, title, year, keywords, abstract. Filters: group, keyword, type. fulltext=True searches PDF text. |
| `find_by_identifier` | A DOI or doi.org link, a key such as Sports:12, a record number, or title words. |
| `get_item` | Every non-empty field, PDF paths (with whether each exists), groups and colour tags. |
| `get_notes` | Notes and research notes for one record, or search them by words. |
| `get_annotations` | Highlights and comments in attached PDFs, with page, text, colour and author. |
| `get_fulltext` | Text of the first attached PDF, 30,000 characters by default. Cached. |
| `list_collections` | Groups with type and item count. Smart groups show their cached members. |
| `collection_items` | Members of one group, by name or group id. |
| `list_tags` | Keywords with counts, or EndNote colour tags with counts (kind=colour). |
| `recent_items` | Newest records by date added or last updated (UTC). |
| `find_duplicates` | Records sharing a DOI, or the same title and year. |
| `list_by_status` | Records whose keywords or Label start with a prefix such as replication or verify. |
| `method_matrix` | Method, Estimator, Sample and Data lines from research notes or custom fields. |
| `get_bibliography` | Formatted bibliography with pandoc and a CSL style such as APA. |
| `export_items` | RIS, BibTeX or CSL-JSON for keys or a group. The file is saved and its text returned. |
| `preview_import` | Checks an RIS file against the library. Writes only new records to a fresh RIS file for File > Import. |
| `check_records` | Flags records that would cite badly in Word: an organisation split across author entries, a missing year, or no author. |

## Common requests

- **Is this paper already in my library?** Ask by DOI or title. **find_by_identifier** returns exact matches first.
- **What did I highlight in this paper?** Give its key. **get_annotations** reads the PDF in place, and nothing is saved.
- **Check an import before I make it.** Give the RIS file. **preview_import** marks each record NEW or DUPLICATE and writes the new ones to one file. You run the import in EndNote with File > Import.
- **Check records before citing.** Give keys or a group to **check_records**. An organisation written as several author entries (the Traveling Library problem) is flagged, so you fix it in EndNote before the Word update.
- **Build a methods table.** Put Method:, Estimator:, Sample: and Data: lines in the research notes, then run **method_matrix** for a group.
- **Format references.** Give keys to **get_bibliography** (for example style apa), or a group to **export_items**.

## Keys and libraries

A key is the library name, a colon and the record number, for example Sports:12. Record numbers are unique only within one library, so the library name is always part of the key. Every tool also takes a library argument, which may be a name or a path. The first library in libraries.json is the default.

```
Sports:12                              record 12 in the Sports library
find_by_identifier  identifier="10.1186/s13102-024-00914-5"  library="Sports"
```

## Limits

- Read only. Adding or changing records and tags needs EndNote itself.
- Results come from the last copy. A change in EndNote shows on the next call after EndNote saves it.
- A library with no Data\sdb\sdb.eni file is an older format. Convert it in EndNote 2025 first: File > Open > Open Library.
- .enlx files are not read. Save or export the library as .enl first.
- Smart groups show the members EndNote last cached, which may be out of date. Group sets were not found in the test libraries.
- Sort order for text is an approximation of EndNote's own. Matching by words and DOIs is exact.
- PDF annotations are read with PyMuPDF. Highlighted text may differ slightly from EndNote's display.

## Planned: testing and security

- **Unit tests:** the parsing and check rules, run without opening a library.
- **Integration tests:** all tools against the live libraries, and the EndNote fields in the Word proposal.
- **User simulations:** the common requests run end to end, with the exact messages a user would see.
- **Security review:** each tool checked for path escape, SQL, pandoc arguments, regex cost and writes.
- **Plan and status:** NEXT_SESSION.md in the server folder.

## Where things are

- **Copies of libraries:** %LOCALAPPDATA%\endnote-mcp\cache\
- **Exports and RIS files:** %LOCALAPPDATA%\endnote-mcp\work\
- **Server, tests and notes:** J:\Claude\endnote-mcp\ (server.py, test_mcp_client.py, README.md, SCHEMA.md)
- **Privacy:** Everything runs on this computer. No library content is sent to an online service.